

TARGET-90

JKBOSE CLASS-11TH

STATISTICS

One Shot

MATHS

Mehraj sir



JKBOSE WALLAH

Topics To Be Covered

- 1 **Introduction to Statistics**
- 2 **Measures of Dispersion** (Mean Deviation about Mean, Median)
Variance
Standard Deviation
- 3 **Related Problems** +

STATISTICS



Meaning

Statistics is the science of **collecting**, **organising**, **presenting**, **analysing** and **interpreting** data.



Types of Data

- Primary Data (Collected first hand)
- Secondary Data (Already available)

Measures of Central Tendency

Mean

$$\bar{x} = \frac{\sum x_i}{n}$$

Median

Arrange in order, find middle value.

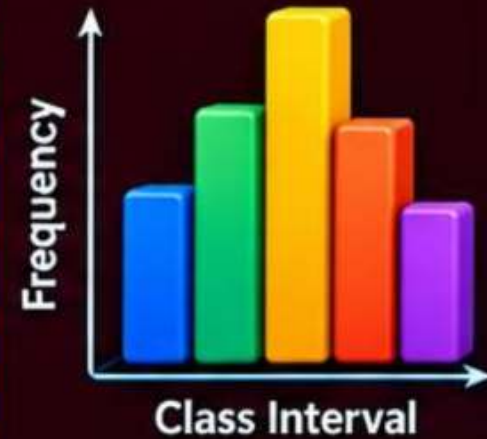
Mode

Most repeated value.

Frequency Distribution

A table showing the **number of times** each value or class occurs.

Class Interval	Frequency
0 – 10	5
10 – 20	8
20 – 30	12
30 – 40	7
40 – 50	3



Graphs

- Bar Graph – Comparison
- Pie Chart – Parts of a whole
- Histogram – Continuous data
- Line Graph – Change over time
- Ogive – Cumulative frequency



Correlation

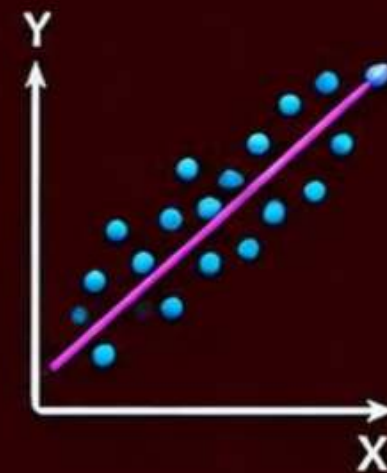
Shows the **relationship** between two variables.

Karl Pearson's Coefficient

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

Range

$$-1 \leq r \leq 1$$



Probability

The **chance** of an **event** occurring.

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

Key Points

- $0 \leq P(E) \leq 1$
- $P(E) = 1$ (Certain event)
- $P(E) = 0$ (Impossible event)



INTRODUCTION

★ **Definition:** The Science of Collecting, organising, Presenting and Interpreting of Data is known as statistics

* Data : The Collection of Numerical figures which represents a Particular information is known as a Data.



Primary Data

- Data collected by an individual or by a group of investigators directly, is called a Primary data.
- Its also called the first hand data.
Example: Data collected by a five students about the height of 50 plants in their school.

Secondary data

- Data collected from a source which already had the information stored, is called a Secondary data
- Its also called the Second hand data.
Example: Data obtained from internet, Books, Magazines etc.

Raw Data

- Initial collection of observations, which is not following any order is called raw data.
Example: The marks obtained by 10 students in a monthly test.

Grouped data

- In order to present data in a more meaningful way, we condense the data into classes or groups. This form of data is called grouped data.

Example:

Height of students	5'0''	5'2''	5'4''	5'6''
Number of students	6	2	5	4

Arrayed data

- Data arranged in increasing order or in decreasing order is said to be an arrayed data.

QUESTION

Primary data can be collected from

- 1 Books ✗
- 2 Historical facts ✗
- 3 Interviews ✓✓
- 4 Magazines ✗

QUESTION

The given data : 5, 8, 7, 6, 4, 2, 9, 7, 3, 1, 2, 4 is

- 1 Raw data ✓
- 2 Arrayed data ✗
- 3 Primary data ✗
- 4 None of these

QUESTION

The weight of 10 students is : 14, 16, 18, 19, 20, 21, 22, 28, 29, 30 The given data corresponds to

- 1 Raw data ✓
- 2 Arrayed data ✓✓
- 3 Primary data ✗
- 4 Secondary data ✗

QUESTION

The data corresponds to

Marks	50	60	89	90	95
Number of Students	4	3	7	6	2

- 1 Raw data ~~X~~
- 2 Arrayed data ~~X~~
- 3 Primary data ~~X~~
- 4 Grouped data ✓✓

Measures of Dispersion

- It is a **Statistical tool** used to describe the spread or variability of a data set.
- It tells us how much the data values differ from the average (mean or median) and from each other.

Mean deviation for ungrouped data

observations: $x_1, x_2, x_3, \dots, x_n$

Mean $\bar{x}$, Median M .

Mean Deviation from $\bar{x}$ is: $M.D(\bar{x}) = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n}$

Mean Deviation from M is: $M.D.(M) = \frac{\sum_{i=1}^n |x_i - M|}{n}$

EXAMPLE



Find the mean deviation about the **mean** for the following data:

6, 7, 10, 12, 13, 4, 8, 12

$$(Sol) \text{ Here } \bar{x} = \frac{6+7+10+12+13+4+8+12}{8}$$

$$\Rightarrow \bar{x} = \frac{72}{8} \Rightarrow \bar{x} = 9$$

$$\text{Therefore } M.D(\bar{x}) = \frac{3+2+1+3+4+5+1+3}{8}$$

$$\Rightarrow M.D(\bar{x}) = \frac{22}{8} \Rightarrow \boxed{M.D(\bar{x}) = 2.75}$$

$$\text{Median } M = \begin{cases} T_{\frac{n+1}{2}} & \text{if } n \text{ is odd} \\ \frac{1}{2} \left(T_{\frac{n}{2}} + T_{\frac{n}{2}+1} \right) & \text{if } n \text{ is even} \end{cases}$$

EXAMPLE



Find the mean deviation about the ^{Median} ~~mean~~ for the following data:

3, 9, 5, 3, 12, 10, 18, 4, 7, 19, 21.

(Sol) The data is shown in the ascending order as follows:

3, 3, 4, 5, 7, 9, 10, 12, 18, 19, 21

Here $n = 11$ (no. of observations), which is odd.

So, Median, $M = T_{\frac{11+1}{2}} = T_6 = 9$.

$$\text{Therefore } M.D(M) = \frac{6+6+5+4+2+0+1+3+9+10+12}{11}$$

$$\Rightarrow M.D(M) = \frac{58}{11} = 5.28.$$

$$\Rightarrow \boxed{M.D(M) = 5.28.}$$

QUESTION (NCERT EXAMPLER)

The mean deviation of data 2, 9, 9, 3, 6, 9, 4 from the mean is

- 1 2.23 ✗
- 2 2.57 ✓✓
- 3 3.23 ✗
- 4 3.57 ✗

QUESTION (NCERT EXAMPLER)

Mean deviation for n observations $x_1, x_2, \dots, x_n$ from their mean $\bar{x}$ is given by

1 $\sum_{i=1}^n (x_i - \bar{x})$

2 $\frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}|$ ✓✓

3 $\sum_{i=1}^n (x_i - \bar{x})^2$

4 $\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ ✓

Mean deviation for grouped data

$$N = \sum_{i=1}^n f_i$$

$$M.D(\bar{x}) = \frac{\sum_{i=1}^n f_i |x_i - \bar{x}|}{N}$$

$$M.D(M) = \frac{\sum_{i=1}^n f_i |x_i - M|}{N}$$

EXAMPLE

Find mean deviation about the **mean** for the following data:

x_i	2	5	6	8	10	12
f_i	2	8	10	7	8	5

(Sol) Here $\bar{x} = \frac{\sum f_i x_i}{N} = \frac{300}{40} = 7.5$.

$$\therefore \text{M.D}(\bar{x}) = \frac{\sum f_i |x_i - \bar{x}|}{N} = \frac{92}{40}$$

$$\Rightarrow \boxed{\text{M.D}(\bar{x}) = 2.3}$$

Hence the result.

x_i	f_i	$f_i x_i$	$x_i - \bar{x}$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
2	2	4	-5.5	5.5	11
5	8	40	-2.5	2.5	20
6	10	60	-1.5	1.5	15
8	7	56	0.5	0.5	3.5
10	8	80	2.5	2.5	20
12	5	60	4.5	4.5	22.5
	40	300			92

EXAMPLE



Find mean deviation about the **median** for the following data:

x_i	3	6	9	12	13	15	21	22
f_i	3	4	5	2	4	5	4	3

(Sol) Here the no. of observations, $N = 30$, which is even

$$\text{So, Median } M = \frac{1}{2} \left(T_{\frac{N}{2}} + T_{\frac{N}{2}+1} \right) = \frac{1}{2} (T_{15} + T_{16}) = \frac{13+13}{2}$$

$$\Rightarrow \boxed{M = 13}$$

$$\therefore \text{Therefore } M.D(M) = \frac{\sum f_i |x_i - M|}{N} = \frac{149}{30}$$

$$\Rightarrow \boxed{M.D(M) = 4.97}$$

Hence the result.

x_i	f_i	cf	$x_i - M$	$ x_i - M $	$f_i x_i - M $
3	3	3	-10	10	30
6	4	7	-7	7	28
9	5	12	-4	4	20
12	2	14	-1	1	2
13	4	18	0	0	0
15	5	23	2	2	10
21	4	27	8	8	32
22	3	30	9	9	27
	30				149

Continuous Frequency Distribution

- A **continuous frequency distribution** is a series in which the data are classified into different class-intervals without gaps along with their **respective frequencies**.

$$\text{Median, } M = l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$$

Class whose Cumulative frequency is equal to $\frac{N}{2}$ or Just greater than $\frac{N}{2}$ is known as a Median class.

* cf is the Cumulative frequency of the class preceeding the Median class.

* $f \rightarrow$ frequency of the Median class.

* $h \rightarrow$ Size of the class.

* $l \rightarrow$ lower limit of the Median class.

EXAMPLE



Find the mean deviation about the mean for the following data:

Marks obtained	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Number of students	2	3	8	14	8	3	2

10-20	2	15
20-30	3	25
30-40	8	35
40-50	14	45
50-60	8	55
60-70	3	65
70-80	2	75

EXAMPLE

Calculate the mean deviation about **median** for the following data:

Class	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	6	7	15	16	4	2

(Sol) Here $\frac{N}{2} = \frac{50}{2} = 25$, So the Median class = 20-30.

$$\Rightarrow \text{Median, } M = l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h = 20 + \left(\frac{25 - 13}{15} \right) \times 10$$

$$\Rightarrow M = 20 + \frac{12}{3} \times 2$$

$$\Rightarrow M = 28$$

$$\text{Therefore } M.D(M) = \frac{508}{50}$$

$$\Rightarrow M.D(M) = 10.16$$

Hence the result.

class	frequency (f_i)	cf	class mark (x_i)	$x_i - M$	$ x_i - M $	$f_i x_i - M $
0-10	6	6	5	-23	23	138
10-20	7	13	15	-13	13	91
20-30	15	28	25	-3	3	45
30-40	16	44	35	7	7	112
40-50	4	48	45	17	17	68
50-60	2	50	55	27	27	54
	50					508

QUESTION



Find the mean deviation about the mean for the data in Exercises 1 and 2.

4, 7, 8, 9, 10, 12, 13, 17

H/w

QUESTION



Find the mean deviation about the mean for the data in Exercises 1 and 2.

13, 17, 16, 14, 11, 13, 10, 16, 11, 18, 12, 17

h/w

QUESTION



Find the mean deviation about the mean for the data in Exercises 5 and 6.

x_i	5	10	15	20	25
f_i	7	4	6	3	5

h/w

QUESTION



Find the mean deviation about the mean for the data in Exercises 5 and 6.

x_i	5	7	9	10	12	15
f_i	8	6	2	2	2	6

H/W

QUESTION



Find the mean deviation about median for the follow data:

Marks	0-10	10-20	20-30	30-40	40-50	50-60
Number of Girls	6	8	14	16	4	2

H/W

Variance and Standard Deviation

* **Definition:** Mean of the Squares of the deviations from the mean is known as Variance.

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}, \quad \sigma^2 = \frac{\sum f_i (x_i - \bar{x})^2}{N}$$

* The Positive Square root of the Variance is known as the Standard Deviation.

$$\sigma^2 = (\text{Mean of } x^2) - (\text{Mean of } \bar{x})^2$$

★ Variance and Standard Deviation of first n natural numbers: $(1, 2, 3, 4, 5, \dots, n)$

$$\text{Here } \sigma^2 = \frac{1^2 + 2^2 + 3^2 + \dots + n^2}{n} - \left(\frac{1 + 2 + 3 + \dots + n}{n} \right)^2$$

$$\Rightarrow \sigma^2 = \frac{n(n+1)(2n+1)}{6n} - \left(\frac{n(n+1)}{2n} \right)^2$$

$$\Rightarrow \sigma^2 = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4}$$

$$\Rightarrow \sigma^2 = \frac{n+1}{2} \left\{ \frac{(2n+1)}{3} - \frac{(n+1)}{2} \right\}$$

$$\Rightarrow \sigma^2 = \left(\frac{n+1}{2} \right) \left\{ \frac{4n+2-3n-3}{6} \right\}$$

$$\Rightarrow \sigma^2 = \frac{(n+1)}{2} \times \frac{(n-1)}{6}$$

$$\Rightarrow \sigma^2 = \frac{n^2-1}{12}$$

Also, Standard deviation $= \sqrt{\frac{n^2-1}{12}}$

QUESTION (NCERT EXAMPLER)

The standard deviation of the data 6, 5, 9, 13, 12, 8, 10 is.

1 $\sqrt{\frac{52}{7}}$

2 $\frac{52}{7}$

3 $\sqrt{6}$

4 6

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

$$* \bar{x} = \frac{6+5+9+13+12+8+10}{7}$$

$$\Rightarrow \bar{x} = \frac{63}{7} \Rightarrow \bar{x} = 9$$

$$\therefore \sigma^2 = \frac{(-3)^2 + (-4)^2 + 0^2 + (4)^2 + (3)^2 + (-1)^2 + (1)^2}{7}$$

$$\Rightarrow \sigma^2 = \frac{9+16+16+9+1+1}{7} = \frac{52}{7}$$

$$\Rightarrow \sigma = \sqrt{\frac{52}{7}}$$

QUESTION (NCERT EXAMPLER)



Standard deviations for first 10 natural numbers is.

1 5.5 ✗

$$S.D. = \sqrt{\frac{n^2 - 1}{12}}$$

2 3.87 ✗

$$\Rightarrow S.D. = \sqrt{\frac{100 - 1}{12}}$$

3 2.97 ✗

$$\Rightarrow S.D. = \sqrt{\frac{99}{12}}$$

4 2.87 ✓✓

$$\Rightarrow S.D. = 2.87$$

EXAMPLE



Find the variance and standard deviation for the following data:

x_i	4	8	11	17	20	24	32
f_i	3	5	9	5	4	3	1

h/w

EXAMPLE



Calculate mean, variance and standard deviation for the following distribution.

Classes	30–40	40–50	50–60	60–70	70–80	80–90	90–100
Frequency	3	7	12	15	8	3	2

f/w

QUESTION



Find the mean and variance for each of the data in Exercises 1 to 5.

6, 7, 10, 12, 13, 4, 8, 12

H/w

QUESTION



Find the mean and variance for each of the data in Exercises 1 to 5.

First n natural numbers

H/W

QUESTION



Find the mean and variance for the following frequency distributions in Exercises 7 to 8.

Classes	0–30	30–60	60–90	90–120	120–150	150–180	180–210
Frequency	3	3	5	10	3	5	2

H/w

QUESTION



Find the mean and variance for the following frequency distributions in Exercises 7 to 8.

Classes	0–10	10–20	20–30	30–40	40–50
Frequency	5	8	15	16	6

H/W

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Aksha De ★★★★★

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Rehana khatun 😊😊😊😊😊

Thank You

$$d_2 = \frac{x_2 - A}{\frac{1}{h_2} \left(\frac{1}{h_2} \right)}$$